Status	Beendet
Begonnen	Freitag, 21. November 2025, 22:48
Abgeschlossen	Freitag, 12. Dezember 2025, 23:59
Dauer	21 Tage 1 Stunde
Punkte	13,00/19,00
Bewertung	**6,84** von 10,00 (**68,42**%)

Frage **1**

Richtig

Erreichte Punkte 1,00 von 1,00

Consider relation students(studentID, degree, courseID) with bitmap indexes on degree and courseID.

Bitmap index on degree:

ISM: 010101

CS: 101010

Bitmap index on courseID:

DBT: 101000

DBTLAB: 010010

ROC: 000100

DW: 000001

Which table matches these bitmap indexes?

Wählen Sie eine Antwort:

studentID	degree	courseID
1	CS	DBT
2	ISM	DBTLAB
3	CS	ROC
4	ISM	DBT
5	CS	DBTLAB
6	ISM	DW

studentID	degree	courseID
1	CS	DBT
2	ISM	DBTLAB
3	CS	DW
4	ISM	DBT
5	ISM	DBTLAB
6	ISM	ROC

studentID	degree	courseID
1	CS	DBT
2	ISM	DBTLAB
3	CS	DBT
4	ISM	ROC
5	CS	DBTLAB
6	ISM	DW

studentID	degree	courseID
1	CS	DBT
2	ISM	DBTLAB
3	CS	ROC
4	ISM	DBT
5	ISM	DBTLAB
6	ISM	DW

Your answer is correct.

Die richtige Antwort ist:

studentID	degree	courseID
1	CS	DBT
2	ISM	DBTLAB
3	CS	DBT
4	ISM	ROC
5	CS	DBTLAB
6	ISM	DW

Frage **2**

Richtig

Erreichte Punkte 1,00 von 1,00

Given the following properties:

Data file: * 10 million tuples * Each tuple requires 2048 bytes

Disk: * Block size: 4096 bytes

Index: * Search key size: 8 bytes * Pointer size: 12 bytes * Index entries do not span blocks

How many blocks are required to store this sparse index?

(Give your answer as a single integer.)

Antwort: 24510

Die richtige Antwort ist: 24510

Frage **3**

Falsch

Erreichte Punkte 0,00 von 1,00

Given the following properties:

Data file: * 10 million tuples * Each tuple requires 512 bytes

Disk: * Block size: 8192 bytes

Index: * Search key size: 4 bytes * Pointer size: 8 bytes * Index entries do not span blocks

How many blocks are required for storing a second-level sparse index if the first-level index is also sparse?

(Give your answer as a single integer.)

Antwort: 121

Die richtige Antwort ist: 2

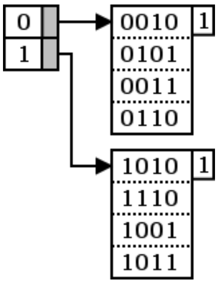
Frage **4**

Richtig

Erreichte Punkte 1,00 von 1,00

Consider the following extensible hash table:

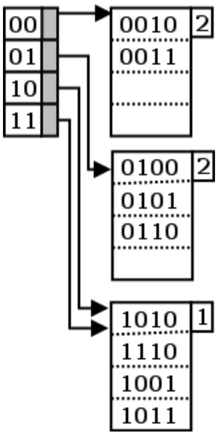
- Block capacity: 4 records
- Current state: $i = 1$

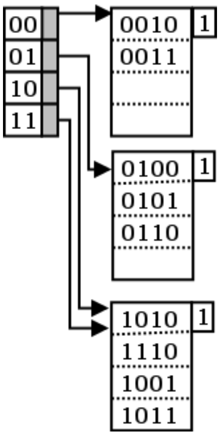


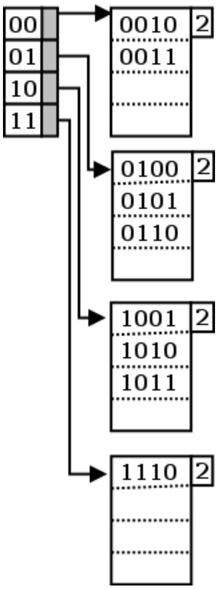
How does the hash table look after inserting key 0100?

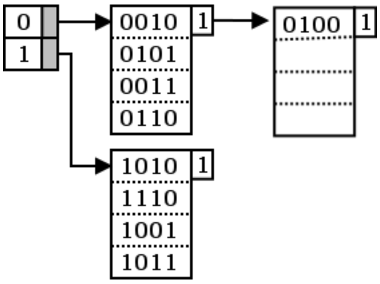
(Hint: See section 14.3.6 in "Database Systems The Complete Book" second edition.)

Wählen Sie eine Antwort:



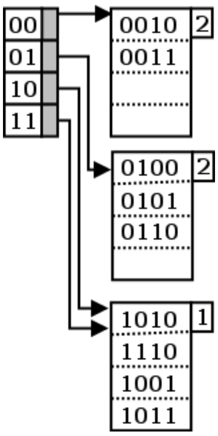






Your answer is correct.

Die richtige Antwort ist:



Frage **5**

Falsch

Erreichte Punkte 0,00 von 1,00

When splitting a block in an extensible hash table and distributing keys into new blocks, one of the new blocks can be empty.

- Wahr
- Falsch ⊗

Die richtige Antwort ist 'Wahr'.

Frage **6**

Richtig

Erreichte Punkte 1,00 von 1,00

Which of the following statements is true?

True	False			
◉	○	In both B and B⁺ Trees, paths from root to leaf have the same length.	✓	
◉	○	Leaves in a B⁺-Tree contain keys and pointers to the block containing the data.	✓	
○	◉	The leaf nodes of both B and B⁺ Trees are connected to the adjacent right leaf node.	✓	
○	◉	Leaves in a B⁺-Tree contain keys and values (i.e., complete tuples with a key).	✓	

In both B and B⁺ Trees, paths from root to leaf have the same length.: True
Leaves in a B⁺-Tree contain keys and pointers to the block containing the data.: True
The leaf nodes of both B and B⁺ Trees are connected to the adjacent right leaf node.: False
Leaves in a B⁺-Tree contain keys and values (i.e., complete tuples with a key).: False

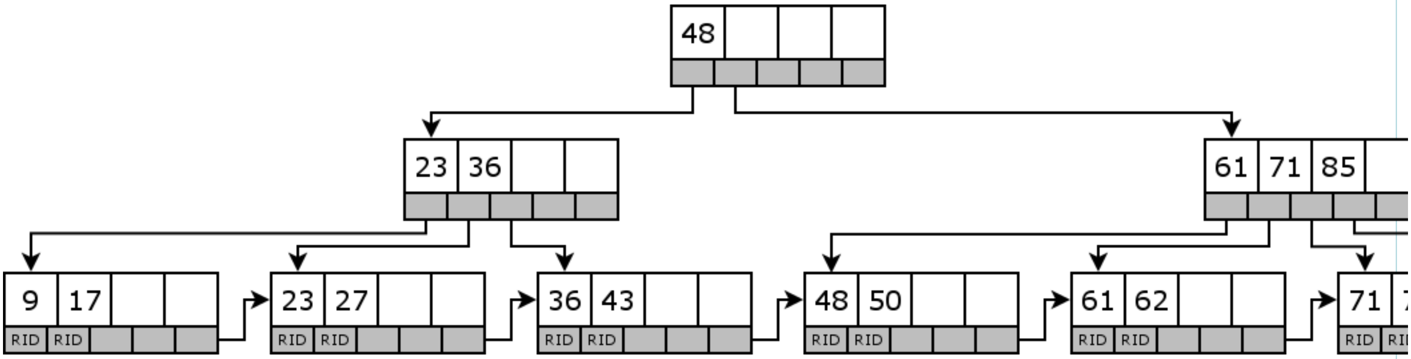
Frage **7**

Richtig

Erreichte Punkte 1,00 von 1,00

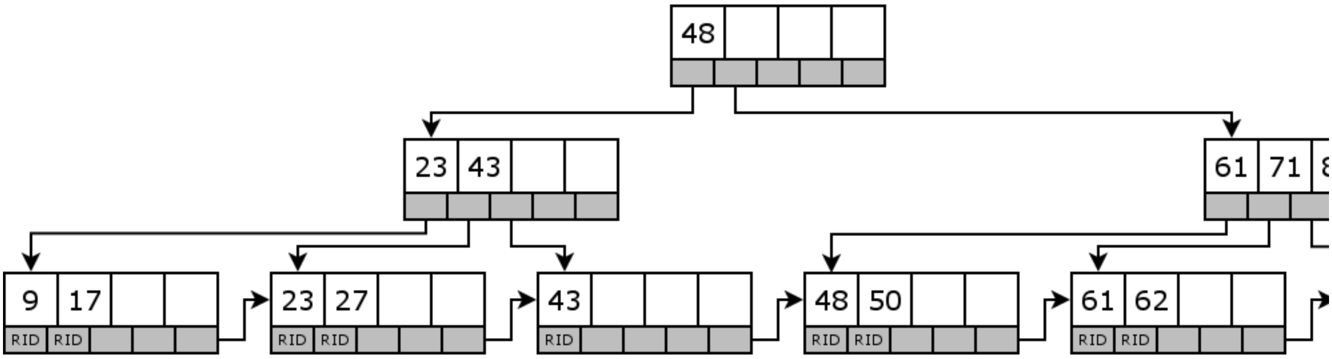
Consider the B⁺ tree below with the following properties:

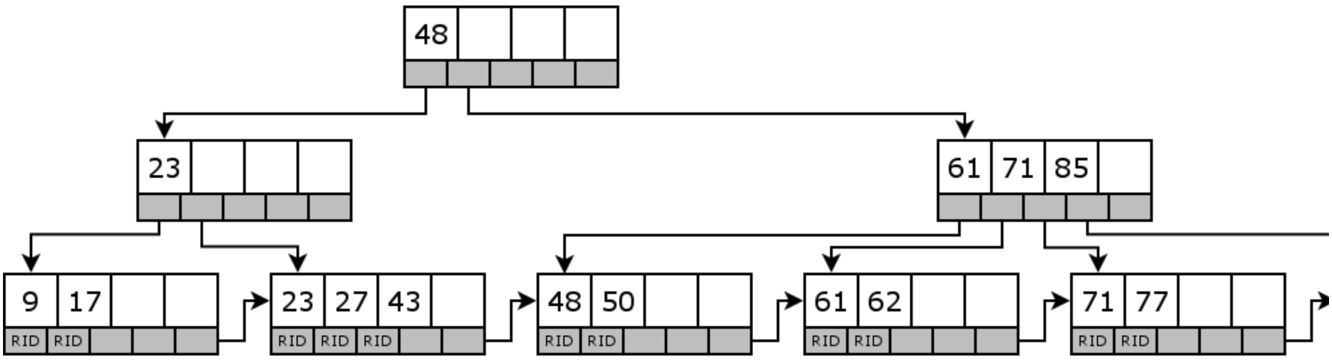
- Each node (except the root) contains $2 \leq k \leq 4$ keys.
- Inner nodes: The keys in the subtree below the pointer p_i are less than the key k_i; the keys in the subtree below the pointer p_{i+1} are greater or equal than the key k_i.
- Delete operations will first try to steal nodes from a direct sibling (first right sibling, then left sibling). If this is not possible, they will merge with a sibling (first right sibling, then left sibling).
- Pointers $p_1, \ldots, p_4$ in leaves point to row IDs. Pointer p_5 in leaves points to the next leaf.
- **Please read section 14.2.6 in "Database Systems The Complete Book" second edition to look for the details of the deletion algorithm when one or more of the above conditions are violated post deletion.**

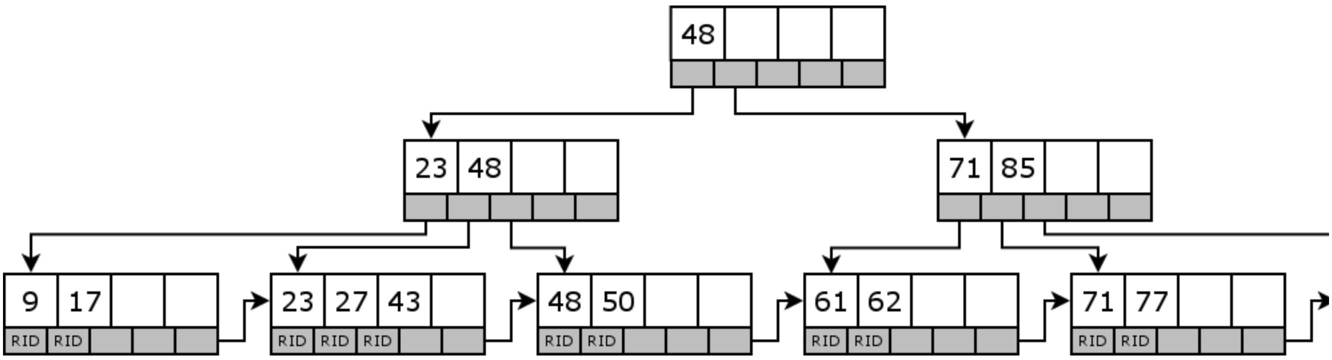


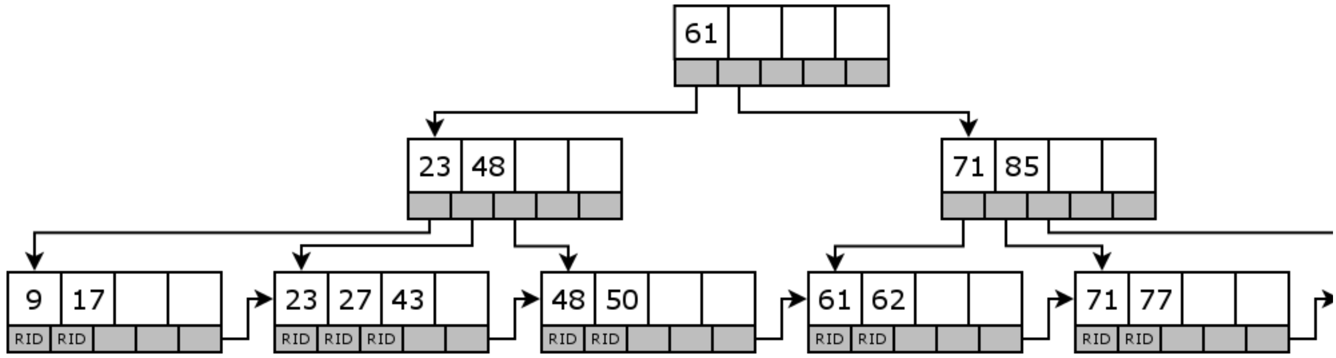
Which of the following choices represents the B⁺ tree after deletion of the key 36?

Wählen Sie eine Antwort:



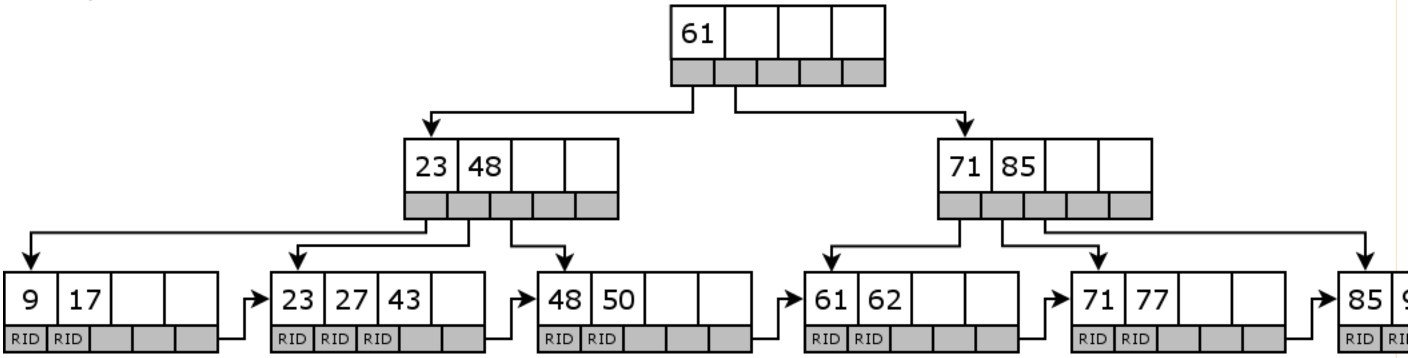






Your answer is correct.

Die richtige Antwort ist:



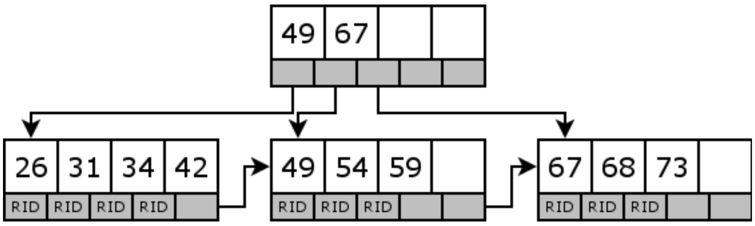
Frage **8**

Teilweise richtig

Erreichte Punkte 0,25 von 1,00

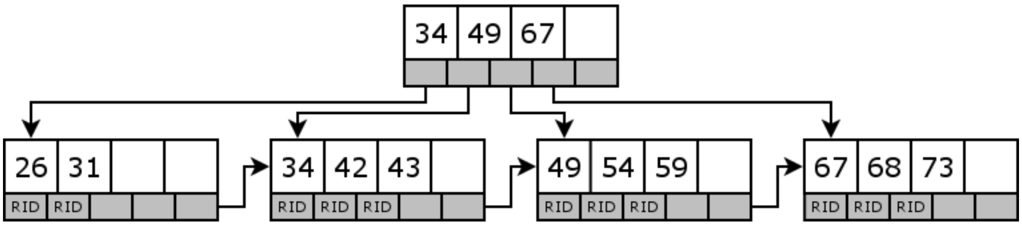
Consider the B⁺ tree below with the following properties:

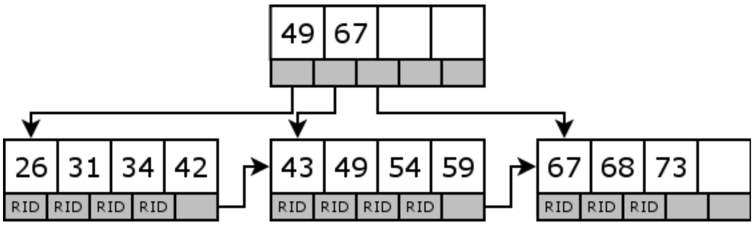
- Each node (except the root) contains $2 \leq k \leq 4$ keys.
- Inner nodes: The keys in the subtree below the pointer p_i are less than the key k_i; the keys in the subtree below the pointer p_{i+1} are greater or equal than the key k_i.
- Insert operations may only trigger node splits but not shifting of keys into bordering leafs.
- When a node is split, the middle key is moved to the parent node.
- Pointers $p_1, ..., p_4$ in leaves point to row IDs. Pointer p_5 in leaves points to the next leaf.

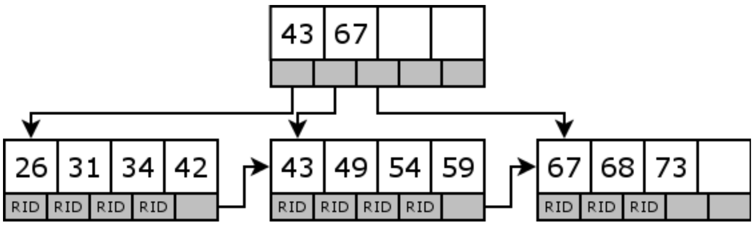


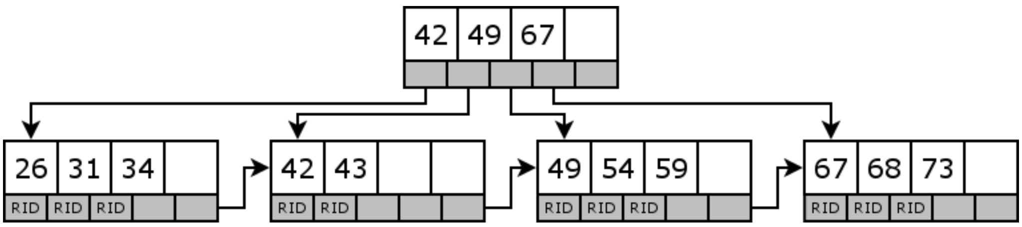
Which of the following choices represents the B⁺-Tree after insertion of the key 43?

Wählen Sie eine Antwort:





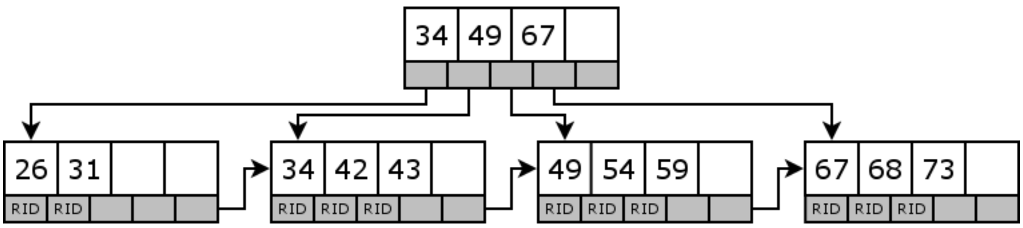




Wrong key moved to the parent root.

Your answer is partially correct.

Die richtige Antwort ist:



Frage **9**

Richtig

Erreichte Punkte 1,00 von 1,00

In a B⁺ tree, all paths from the root to a leaf have the same length.

- Wahr
- Falsch

Die richtige Antwort ist 'Wahr'.

Frage **10**

Falsch

Erreichte Punkte 0,00 von 1,00

Given the following properties of the data file, the disk, and index structures.

Data file:

- The data file contains 10 million tuples.
- Each tuple requires 512 bytes on disk.

Disk properties:

- The size of a disk block is 8192 bytes.

Index properties:

- The size of a search key is 4 bytes.
- The size of a pointer to a block is 8 bytes.
- Index entries do not span blocks on disk.

How many blocks are required to store this dense index?

(Give your answer as a single integer.)

Antwort: 14664

Die richtige Antwort ist: 14663

Frage **11**

Richtig

Erreichte Punkte 1,00 von 1,00

Given fact table Clicks(<u>User</u>, <u>Session</u>, <u>Page</u>, Position) with a grid file having the following partitions:

- User: 4 partitions
- Session: 3 partitions
- Page: 9 partitions

After ingesting more data, one partition of the Page dimension is split to accommodate the extra data.

How many buckets now exist in total?

Antwort: 120

Die richtige Antwort ist: 120

Frage **12**

Teilweise richtig

Erreichte Punkte 0,75 von 1,00

Which of the following statements are true?

True	False			
◉	○	A linear hash table searches the bucket array using the least significant bits.	✓	
○	◉	An extensible hash table is optimized for an even distribution of data across blocks.	✓	
○	◉	An extensible hash table has a constant search cost.	✗	
○	◉	A linear hash table searches the bucket array using the most significant bits.	✓	

A linear hash table searches the bucket array using the least significant bits.: True
An extensible hash table is optimized for an even distribution of data across blocks.: False
An extensible hash table has a constant search cost.: True
A linear hash table searches the bucket array using the most significant bits.: False

Frage **13**

Richtig

Erreichte Punkte 1,00 von 1,00

In a highly selective query, only a few elements qualify under the selection; most elements are discarded.

(Note: Selectivity refers to the property of the index or query.)

◉ Wahr ✓

○ Falsch

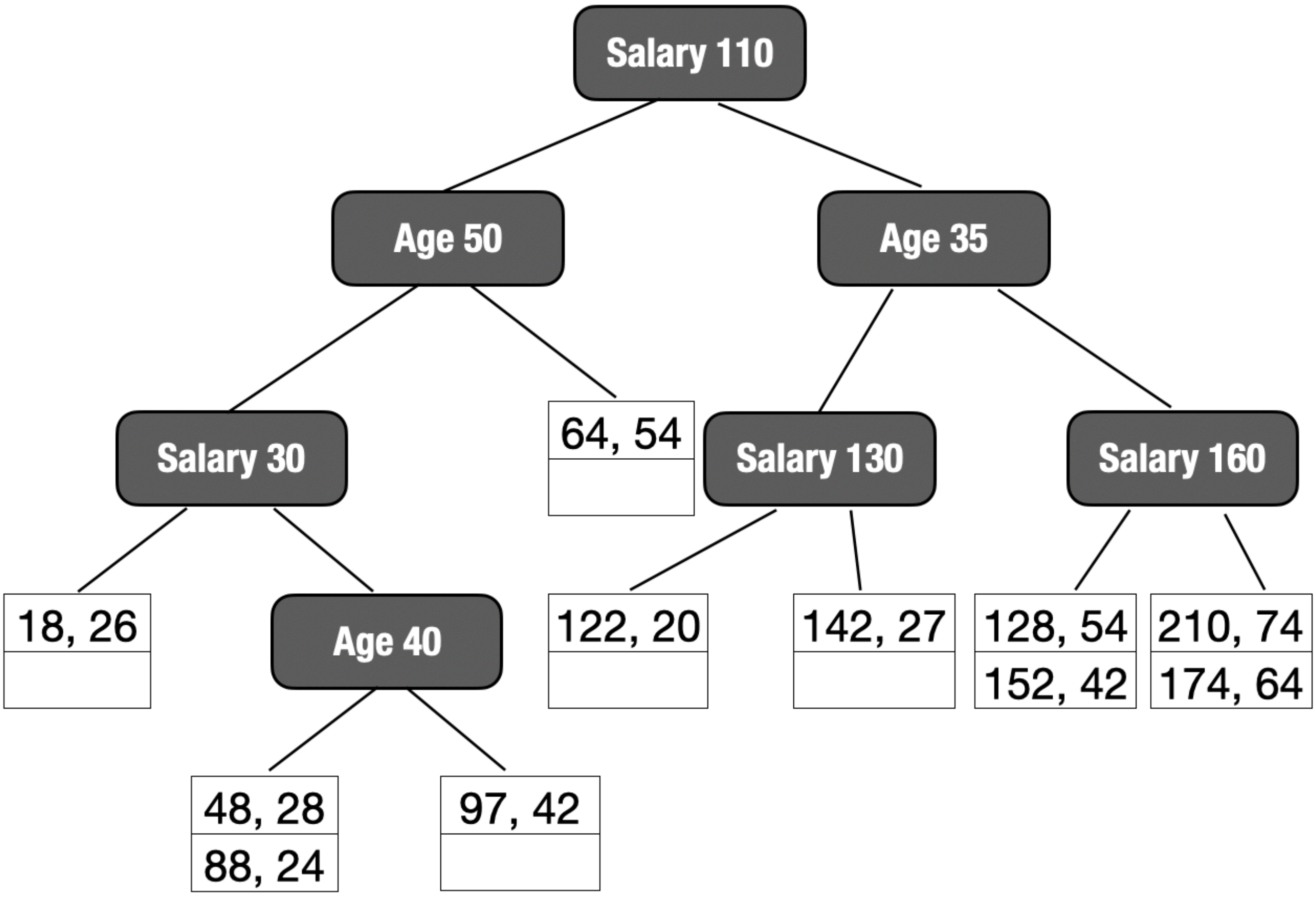
Die richtige Antwort ist 'Wahr'.

Frage **14**

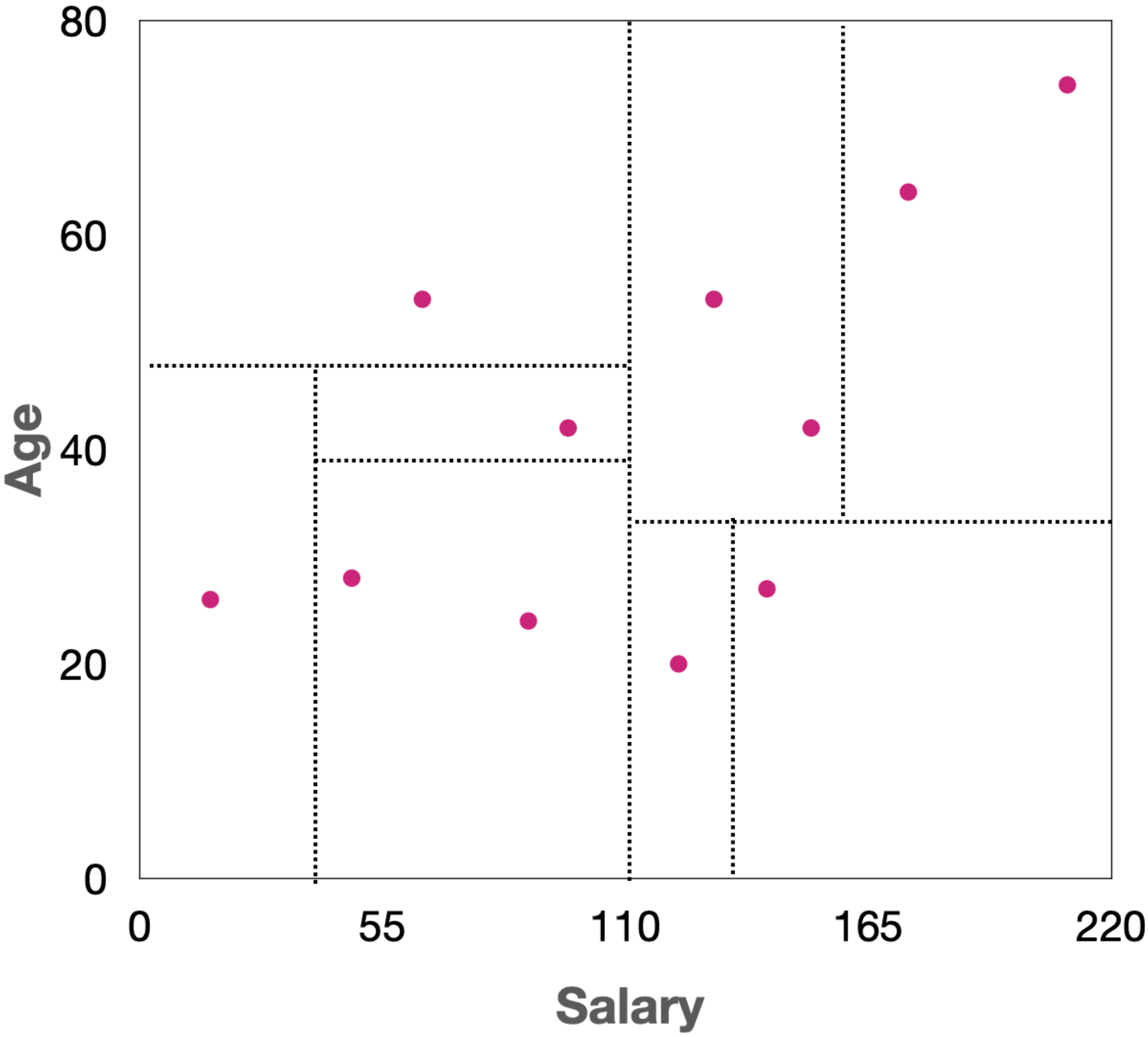
Falsch

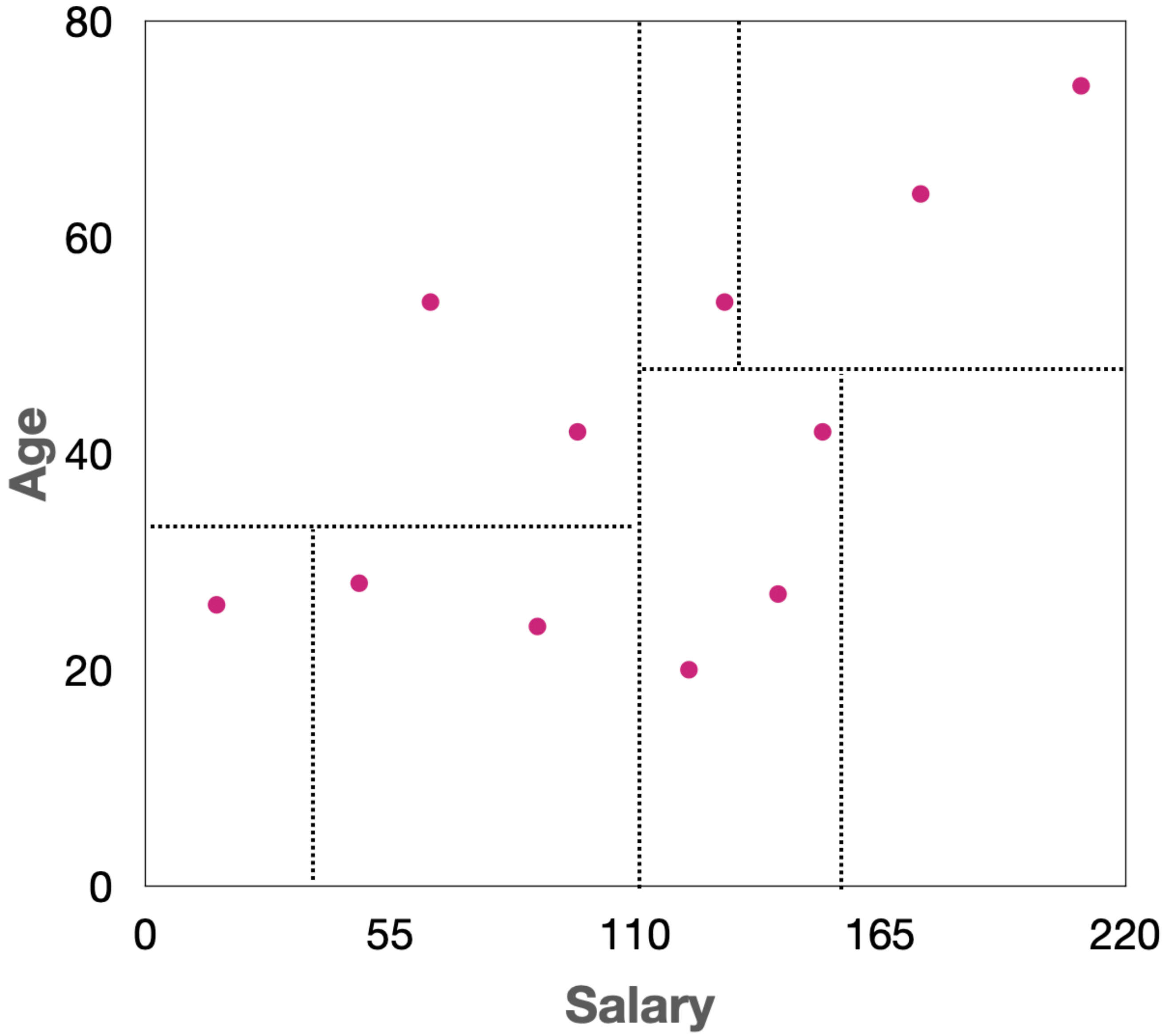
Erreichte Punkte 0,00 von 1,00

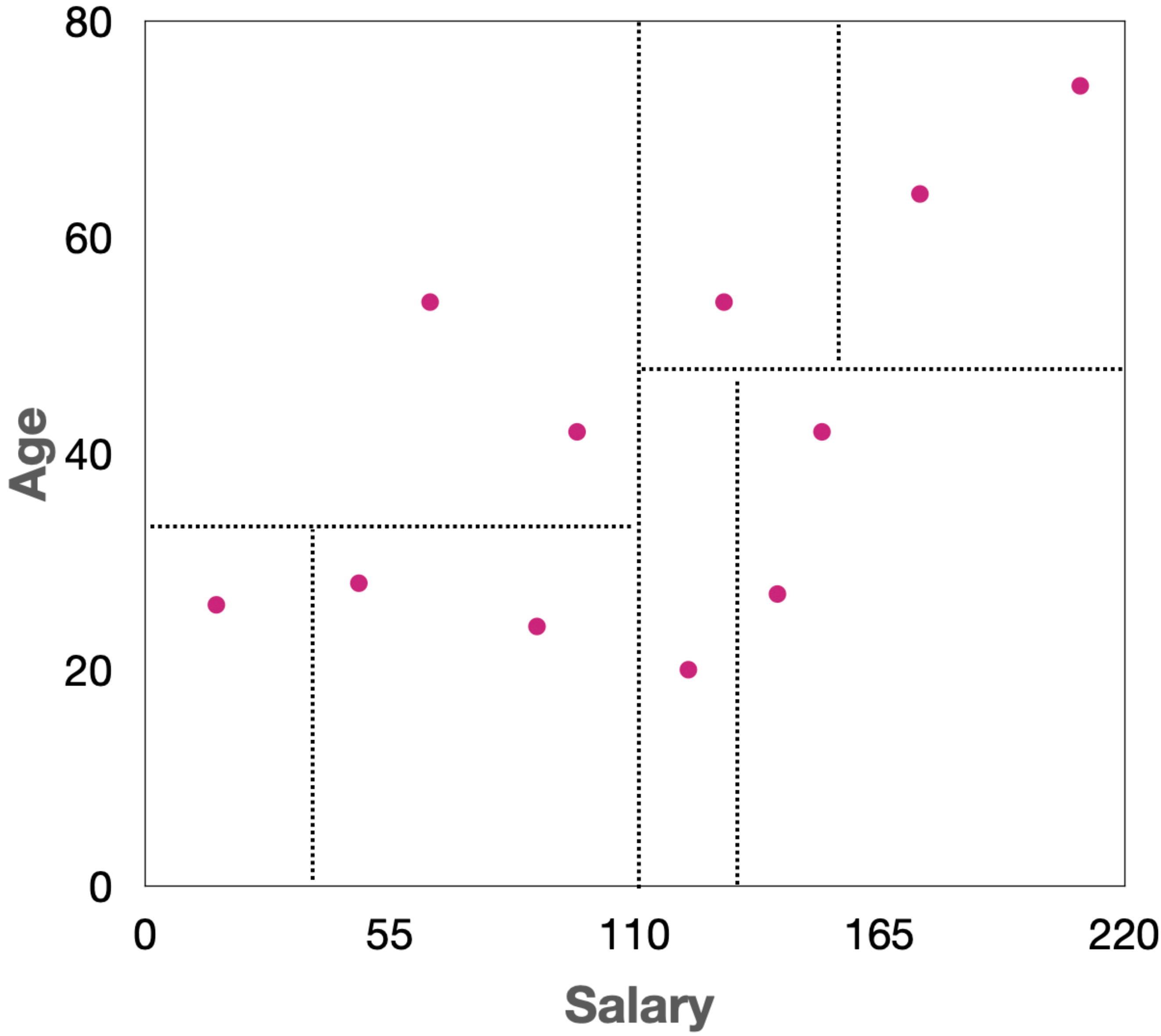
Given the following kd-tree, choose the corresponding partitions.

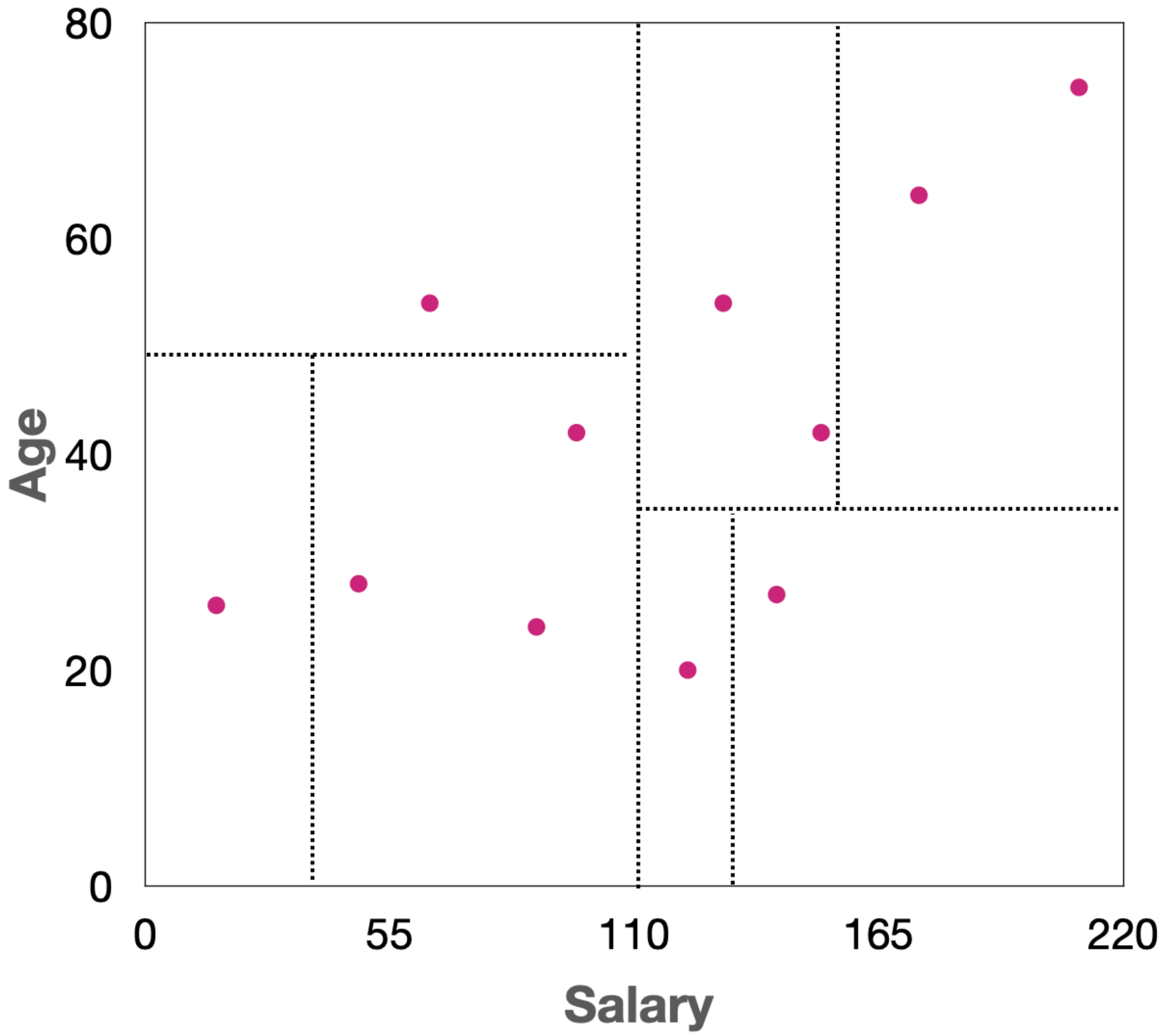


Wählen Sie eine Antwort:


80
60
40
20
0
Age
0
55
110
165
220
Salary

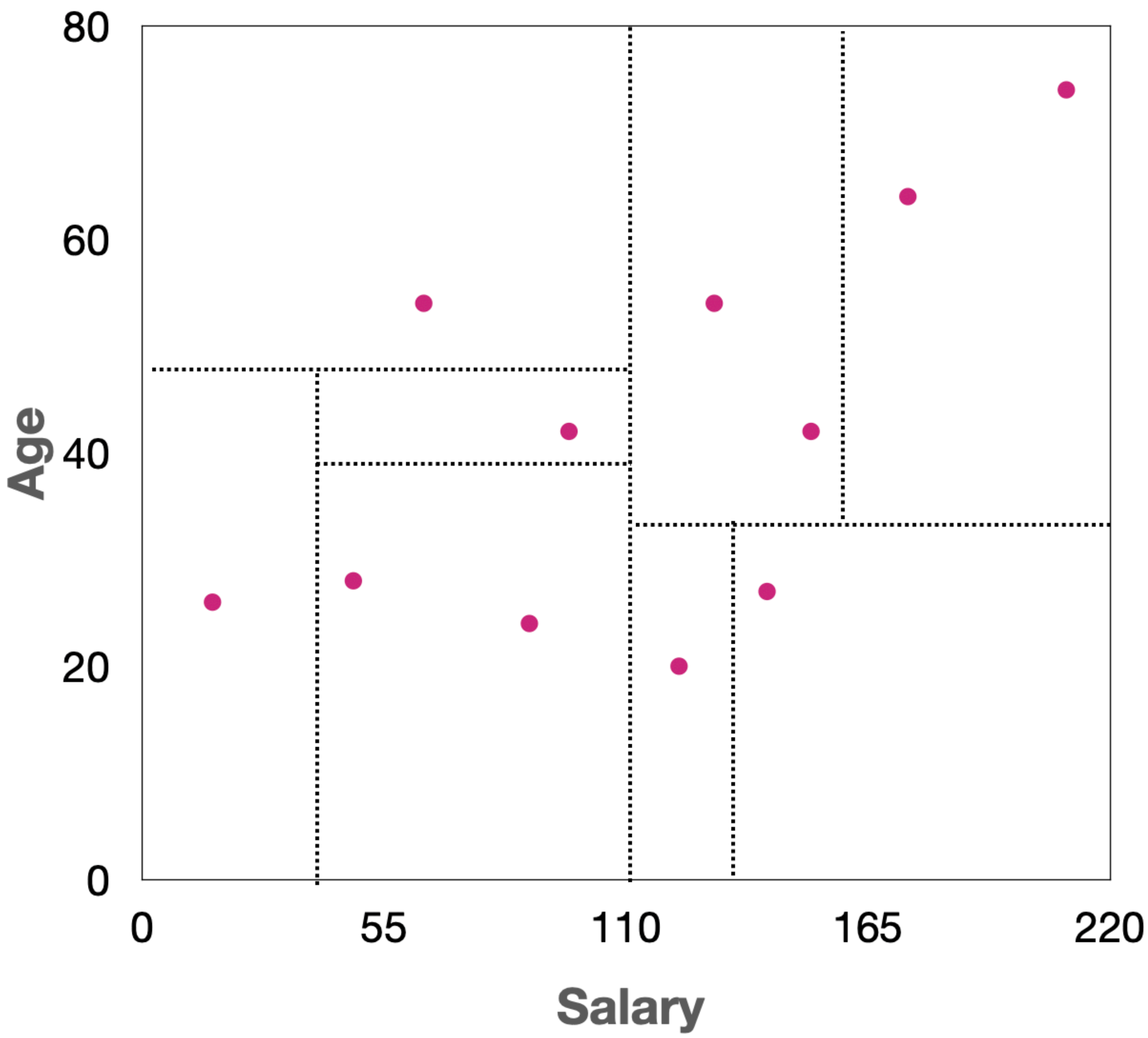

80
60
40
20
0
Age
0
55
110
165
220
Salary


80
60
40
20
0
Age
0
55
110
165
220
Salary



Your answer is incorrect.

Die richtige Antwort ist:



Frage **15**

Richtig

Erreichte Punkte 1,00 von 1,00

When adding a new block to extend a linear hash table, we always have to rehash one existing bucket.

- ◉ Wahr ✓
- ○ Falsch

Die richtige Antwort ist 'Wahr'.

Frage **16**

Falsch

Erreichte Punkte 0,00 von 1,00

kd-trees branch on different indexed attributes at the same level.

- Wahr
- Falsch

False. kd-trees branch on different indexed attributes at different levels.

Die richtige Antwort ist 'Falsch'.

Frage **17**

Richtig

Erreichte Punkte 1,00 von 1,00

Consider relation Customer(Age, PostalCode, City, TotalSpent) with a kd-tree index on the primary key.

The following query:

```
select avg(TotalSpent) from Customer
where Age between 45 and 60 and
      PostalCode between 10629 and 13581
```

is an example of:

Wählen Sie eine Antwort:

- GIS Query
- Point Query
- Partial-Match Query
- Range Query

Your answer is correct.

This query specifies a range of values for two dimensions (Age and PostalCode), making it a range query.

Die richtige Antwort ist: Range Query

Frage **18**

Richtig

Erreichte Punkte 1,00 von 1,00

Consider relation R(a, b, c, d) stored using partitioned hashing:

- Total buckets: 65536 (16-bit hash function)
- Bit allocation: 4 bits for a, 5 bits for b, 3 bits for c, 4 bits for d

How many buckets must be checked for a partial match query on a?

Example:

```
SELECT *
FROM R
WHERE a = 'y';
```

Antwort: 4096

Die richtige Antwort ist: 4096

Frage **19**

Richtig

Erreichte Punkte 1,00 von 1,00

A sparse index does not have an entry for every row in the table.

- (x) Wahr
- ( ) Falsch

Die richtige Antwort ist 'Wahr'.